



BEST PRACTICE SERIES

**Using VIMS® to Determine Air
Filter Replacement Intervals**

Maintenance and Repair Application Component Renewal (CRC) Component Life Management MARC Management

Using VIMS® to Determine Air Filter Replacement Intervals 0

1.0 Introduction 1

2.0 Best Practice Description..... 1

3.0 Implementation Steps 3

4.0 Benefits..... 3

5.0 Resources Required 3

6.0 Supporting Attachments / References 4

7.0 Related Best Practices 4

8.0 Acknowledgements..... 4

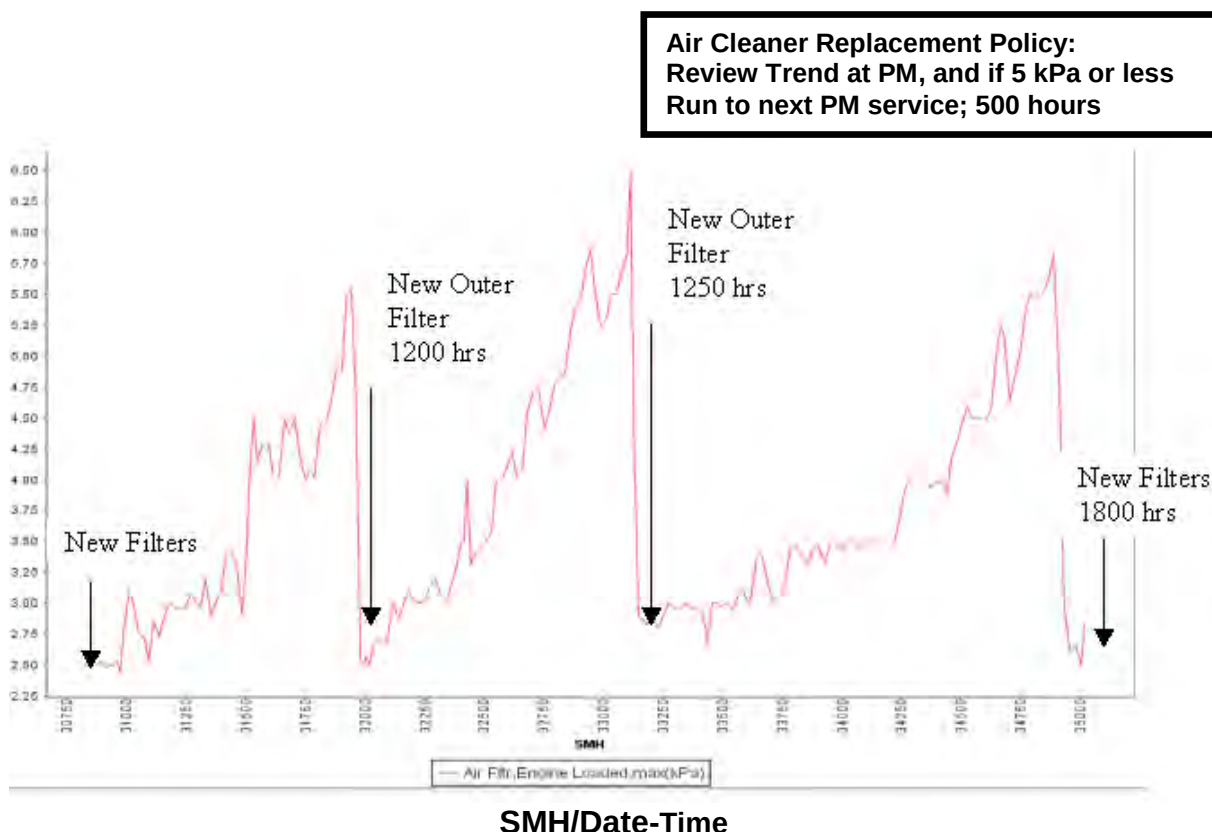
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1.0 Introduction

Typically, a site replaces filters at a fixed number of hours or PM interval. Using VIMS[®], however, it may be possible to manage and optimize air filter service based on condition rather than a predetermined hour interval. This Best Practice discusses the process of Trending Air Filter Restriction using VIMS[®] data and managing change out routines to maximize air filter life based on their condition rather than elapsed time. While some particularly dusty sites may require filter replacement before PM's, many can benefit from the economy of servicing filters only once used up. Implementation of this practice coincides directly with the steps outlined in the Best Practice *Utilizing VIMS[®] Data for Effective Planned Maintenance*. That Best Practice contains the steps for obtaining the necessary VIMS[®] data to make filter replacement decisions.

2.0 Best Practice Description

A new air filter typically causes a restriction of about 2 – 2.5 kPa (8-10 in. H₂O). As the machine is run, dust collects in the filter and increases the restriction pressure. Engine performance begins to decline at a restriction of 6 kPa (24 in. H₂O). Rather than waiting for a predetermined time period to pass, use the air filter restriction data in VIMS[®] to determine a need for filter replacement. For instance, filters could be serviced when restriction pressure reaches 5 kPa (20 in. H₂O).



This graph shows air filter restriction growing with time. Filter replacement is seen at each drop in pressure after maximizing at 6 kPa. Note that the filter change intervals are not exactly the same— filters were changed on condition rather than time.

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Using VIMS[®] to Determine Air Filter Replacement
Intervals

DATE
1/12/2015

CHG
NO
02

NUMBER
0107-2.15-1059

Knowledge of air filter life under particular site conditions will need to be developed to determine whether air filter extensions are possible on the site.

While VIMS® reports air filter restriction at less than 5.0 kPa (24 in. H₂O), no action is required.

When restriction surpasses 5.0 kPa (24 in. H₂O), or if filters cannot wait until the next PM, filters should be serviced.

The Recommended Trend Analysis Period is several days before PM service is due, using the last 250 hours of recorded data. Analyze the rate of change in air filter restriction to serve as a guide to how fast the restriction is occurring.

If air filters will last only one PM service and are expected to plug to the point of engine derate before the next PM service, then their life cannot be extended unless efficient changing of air filters in the field is possible. However, if the filters will run successfully to at least the next PM service, plan to resample several days prior to the PM and again assess whether they can be extended. Repeat this analysis until a point is reached where the filters will not reach the next PM without engine derating.

2.1 Management

An appropriate system for marking filters and recording when filters are changed in the work order must be implemented in order to avoid the cost of unnecessarily changing good filters and to help extend filter life. A Condition Monitoring (CM) analyst or field mechanic should be able to quickly identify when the filter was last serviced in order to avoid servicing new filters should an event arise in the field. Tracking filter changes may also lead to increased inner filter life when one set of inner filters is used with three sets of outer filters.

Air filters actually become more efficient as they begin to become plugged. As they trap particles, some of the smaller holes naturally found in filter media become plugged, preventing other particles from passing through. Extending filter life extends the period that the filter is used when it is more efficient. One of the goals of filter management is to extend filter life, and then to replace the filter before there is any decline in engine performance.

The air filter management process should begin when all new filters have been installed in a machine. When the filters are installed, the inner and outer filter should be marked with the date and SMH, and the information should be recorded in a log available to the onsite CM analyst. This information needs to be accurate and visible to all parties involved in managing air filter life (Planning, Field Repairs, PM techs, Reliability Engineer, CM Analyst, etc.).

When restriction data shows the need to replace a filter, the outer filter should be replaced. The outer filter should again be marked with the date SMH and the inner filter should have an X placed on it. The goal should be to use 3 outer filters to every inner filter, so first change = 1"X", second change = 2"X's", third time = new inner filter.

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Using VIMS® to Determine Air Filter Replacement Intervals	DATE 1/12/2015	CHG NO 02	NUMBER 0107-2.15-1059
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3.0 Implementation Steps

Guidelines for Implementing a CM Based Filter Management Process:

- The process should be implemented on 3-5 trucks. The Best Practice value can be measured before the process is applied to the entire fleet.
- Start the management process with all new filters (inner and outer). Starting the process on a machine with a new or rebuilt engine is also beneficial.
- Gather the VIMS® data prior to the PM. Collecting data at pre-determined intervals will provide consistent results.
- Analyze the VIMS® data. A record should be kept of how much the air restriction drops when a new outer filter is replaced. It should drop to almost the same level as all new filters (slightly higher). If the drop isn't this large it may indicate a need to replace the inner filter as well.
- Make decision based on the data VIMS® provides. The onsite CM analyst should be able to use the log determine if air filter restriction is a possible cause of a low power complaint. This will help avoid changing a good air filter when there is a different root cause.

4.0 Benefits

- Saves time and resources by making maintenance decisions based on the VIMS® data instead of performing maintenance at standard time intervals.
- For an actual fleet of ten 793's over 6 months:
 - o Changing air filters based on time: 268 outer filters (67 sets) at \$800 per set
 - Total parts cost: \$ 53,600
 - o Changing air filters based on VIMS® data: 112 outer filters (28 sets)
 - Total Parts cost \$22,400
 - o Using VIMS® saved \$31,200

5.0 Resources Required

The major resource required to implement this Best Practice will be the people and the time dedicated to gathering and analyzing the data collected from the machine. The VIMS® data will be collected from the machine using the VIMSpC® software. VIMS® Application training may also be required if the users are not familiar with the analysis of VIMS® data.

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Using VIMS® to Determine Air Filter Replacement Intervals	DATE 1/12/2015	CHG NO 02	NUMBER 0107-2.15-1059

6.0 Supporting Attachments / References

References:

VIMSpC® subscription numbers: JERD2137 (Dealer Site License)
 JERD2138 (Dealer Annual Subscription)
 JERD2175 (Customer Annual Subscription)

VIMS® Application Guide SELD7001

Attachments:

VIMS® PM Planner.doc Template for recording VIMS® data before PM's

7.0 Related Best Practices

[1006-2.3-1027 Utilizing VIMS® Data for Effective Planned Maintenance](#)

8.0 Acknowledgements

This Best Practice was written by:

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Using VIMS® to Determine Air Filter Replacement Intervals	DATE 1/12/2015	CHG NO 02	NUMBER 0107-2.15-1059
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